

NovaShyld Cybersecurity Analyst – Week 1 Evidence

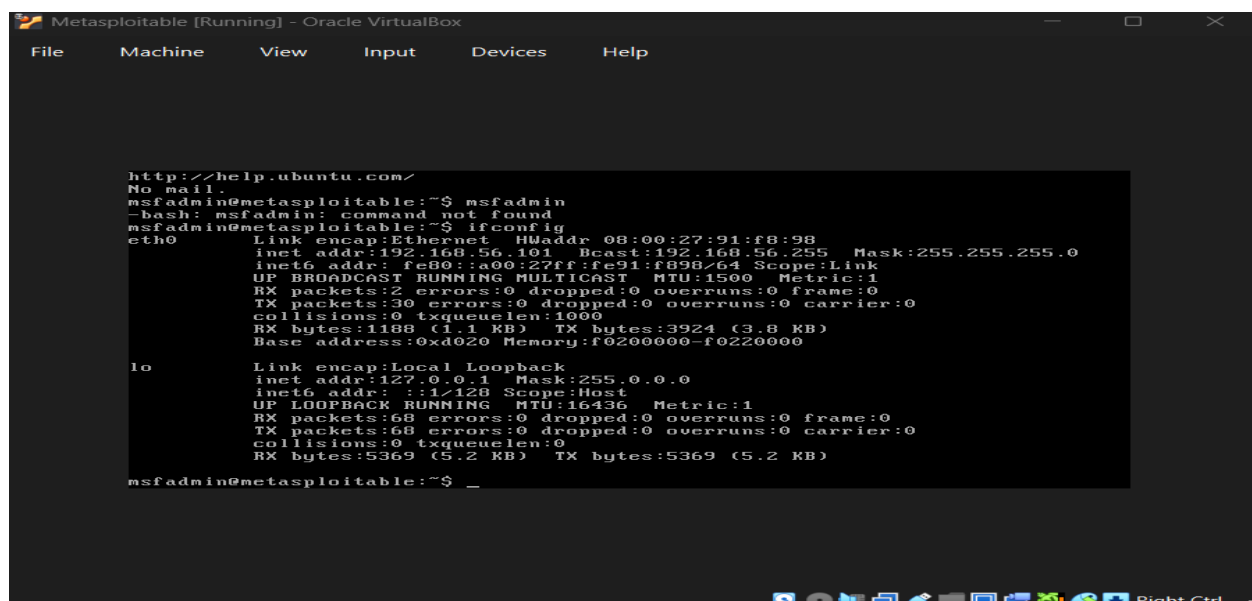
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Kali + Metasploitable

The purpose of this Week 1 task was to set up a safe penetration testing lab using Kali Linux (attacker) and Metasploitable (victim). The goal was to confirm connectivity between two machines, configure a private network adaptor path, and initialize vulnerable web applications platform database. These steps are the foundation of ethical hacking: before discovering or exploiting vulnerabilities, you must first build an isolated sandbox environment to operate responsibly. This report provides screenshots and explanations so that even a first-time learner can follow along.

Task 2: Lab Setup and Security Environment Configuration

A screenshot of a virtual machine window titled "Metasploitable [Running] - Oracle VirtualBox". The window shows a terminal interface with the following text:

```
http://help.ubuntu.com/
msfadmin@metasploitable:~$ msfadmin
-bash: msfadmin: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:91:f8:98
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe91:f898/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2 errors:0 dropped:0 overruns:0 frame:0
          TX packets:30 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1188 (1.1 KB)  TX bytes:3924 (3.8 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:68 errors:0 dropped:0 overruns:0 frame:0
          TX packets:68 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:5369 (5.2 KB)  TX bytes:5369 (5.2 KB)

msfadmin@metasploitable:~$ _
```

Figure 1: Observe network interface configuration

- Open the Metasploitable victim virtual machine console window
- Type the following on the Metasploitable terminal: ipconfig

- This command maps out active adapters, display system MAC details and identifies the core server location
- This is important because it reveals the exact target IP address 192.168.56.101 needed to establish a connection baseline

```

kali@kali: ~
Session Actions Edit View Help
Sorry, try again.
[sudo] password for kali:
sudo: 1 incorrect password attempt

(kali@kali)-[~]
$ sudo systemctl restart NetworkManager
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:

(kali@kali)-[~]
$ ping -c 4 192.168.56.101
ping: connect: Network is unreachable

(kali@kali)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=33.0 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=9.52 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=6.43 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=8.30 ms

— 192.168.56.101 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3467ms
rtt min/avg/max/mdev = 6.427/14.300/32.957/10.827 ms

(kali@kali)-[~]
$

```

Figure 2: Verify network adapter connectivity

- Open the terminal application path inside the running Kali Linux desktop system
- Type the following on the Kali terminal panel: `ping -c 4 192.168.56.101`
- This sends automated testing data packets across the host-only virtual adapter wire link
- This is important because receiving zero percent packet loss confirms that the attacker and victim hosts can fully communicate

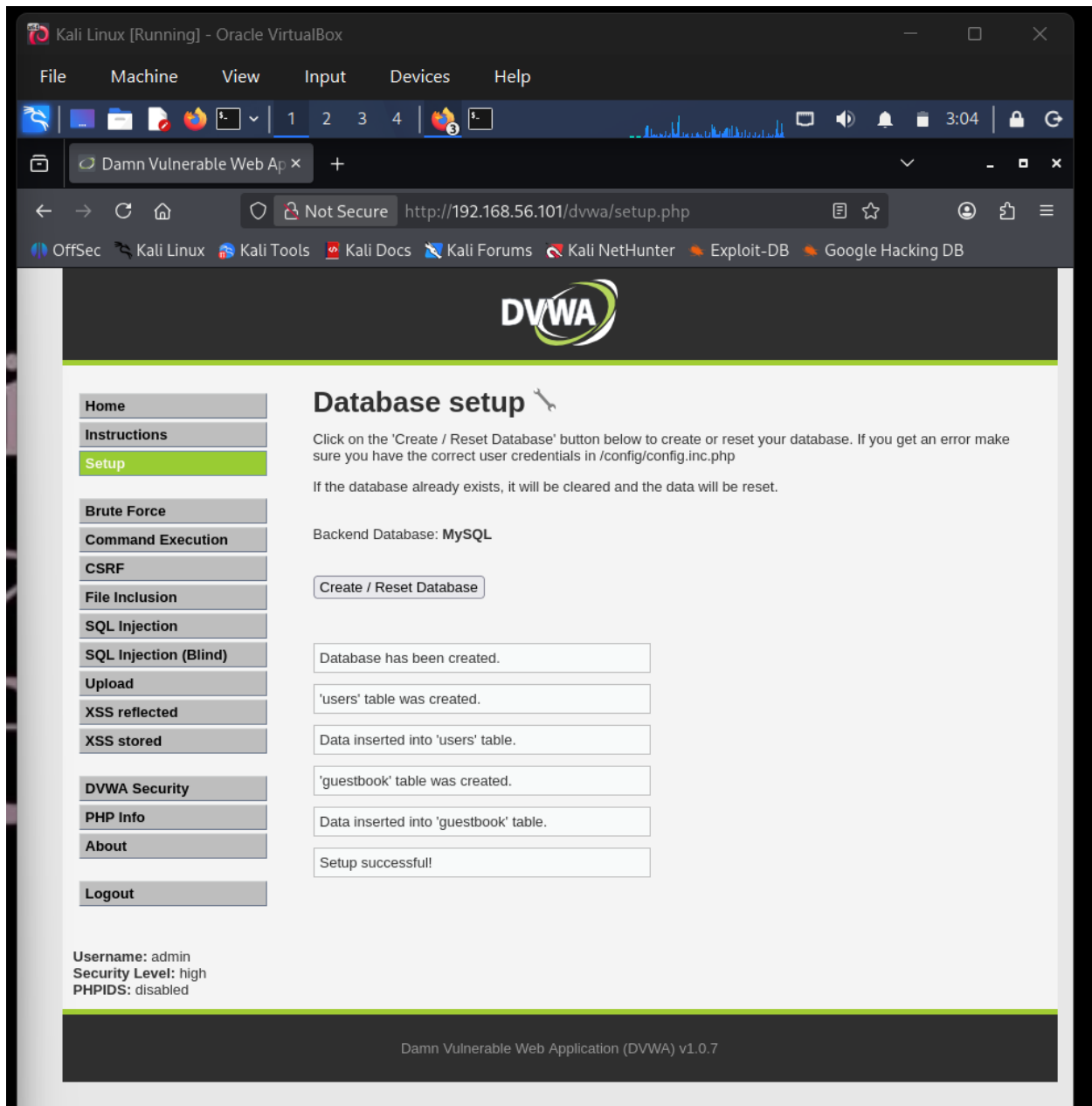


Figure 3: Initialize vulnerable web platform

- Open the Firefox web browser layout window inside the Kali Linux environment
- Type the following URL link into the browser navigation interface bar: <https://192.168.56.101>
- Click the button labeled “Create / Reset Database” to populate the background server tables
- This is important because it builds the underlying data frameworks needed to test web exploitation field successfully

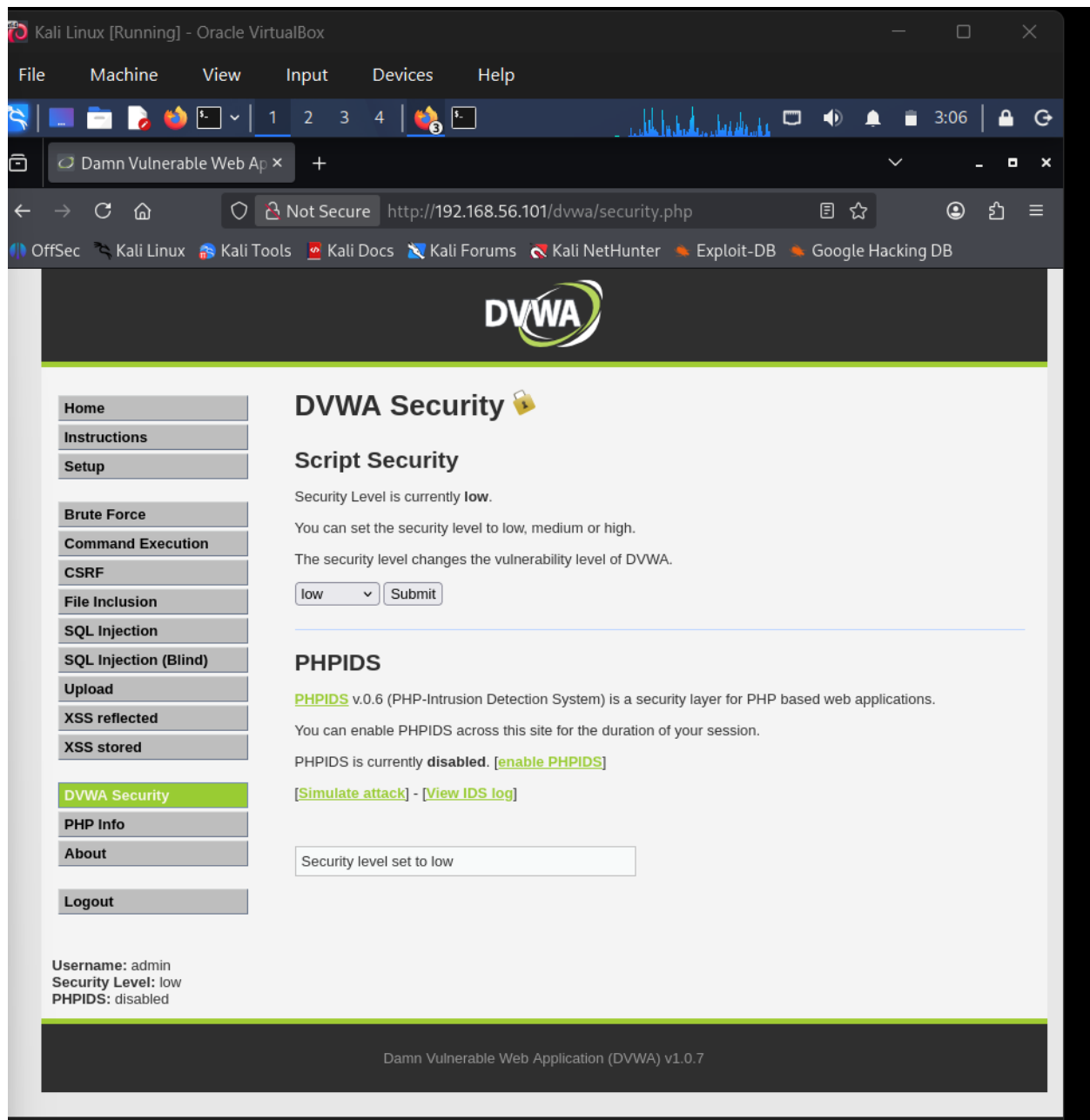


Figure 4: Adjust target configuration parameters

- Click on the DVWA Security setting button located on the left-hand navigation list
- Toggle the script execution drop down field option configuration down to low
- Click the “submit” button to lock the new session parameters into the application memory block
- This is important because changing the security level to low strips away advanced input filters so you can test raw flaws

Findings

- Kali Linux successfully established an internal network communication corridor with the Metasploitable target machine without using outside internet routers
- The victim environment host properly assigned its network identity to address location 192.168.56.101 under the host -only adapter bracket
- The web server daemon listening on port 8 successfully answered the browser request and delivered the target application package
- The backend application environment script profile was updated to an unrestricted testing structure showing that the system state changed successfully

Conclusion

This week's tasks were successfully completed, with both the network setup and application platform initialization showing clear evidence of a secure testing sandbox. The findings show that the lab setup is working as intended, with isolated services accessible for further deeper analysis. By documenting the commands, outputs, and screenshots, the evidence demonstrates completion of Week 1 objectives and provides a strong foundation for deeper exploitation and vulnerability testing in the upcoming weeks.